VECTOR

SFD2 Integration with Vector MICROSAR 4 Stack

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Restrictions Customer Confidential – VAG only

Abstract Guidelines for integrating SFD2 component with Vector MICROSAR 4 Stack

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1 Integration of SFD2 with Vector MICROSAR 4 Stack

This document helps with the integration of the SFD2 component from Elektrobit on the Vector MICROSAR 4 Stack. Please also read the references [1] and [2].

1.1 Adding SFD to a Davinci Configurator Project

For adding the SFD component to an existing Davinci Configurator Pro based project, the application note [1] can be used. Following are few hints to make this process easier:

1. Step 1 – Install Tresos Plugin in [1] is necessary because SFD is not delivered as part of the SIP.
2. Two separate instances of the external generator(`tresos_cmd.bat`) need to be added to the Davinci Project settings. In Step 9, the following has been used as parameters for the generation of SFD component:

   For SFD SWC Descriptions (*.arxml):

   ```
   - Dtarget=<target> -Dderivate=<derivative> legacy make generate_swcd
   $(SipRootPath)\DaVinciConfigurator\Generators\Sfd\Sfd_xslt_Result.arxml@
   asc:4.3.0 -o <OutputFolderPath> -n Sfd -n Csm -n Dcm -g <Sfd Module-id>
   ```

   Example

   ```
   -Dtarget=WINDOWS -Dderivate=Win32x86 legacy make generate_swcd
   $(SipRootPath)\DaVinciConfigurator\Generators\Sfd\Sfd_xslt_Result.
   arxml@asc:4.3.0 -o $(GenDataVTTFolder) -n Sfd -n Csm -n Dcm -g
   Sfd_TS_TxDxM2I12R0
   ```

   For SFD configuration code (*.c and *.h):

   ```
   -Dtarget=<target> -Dderivate=<derivative> legacy generate
   $(SipRootPath)\DaVinciConfigurator\Generators\Sfd\Sfd_xslt_Result.arxml@
   asc:4.3.0 -o <OutputFolderPath> -n Sfd -n Csm -n Dcm -g <Sfd Module-id>
   ```

   Example

   ```
   -Dtarget=WINDOWS -Dderivate=Win32x86 legacy generate
   $(SipRootPath)\DaVinciConfigurator\Generators\Sfd\Sfd_xslt_Result.
   arxml@asc:4.3.0 -o $(GenDataVTTFolder) -n Sfd -n Csm -n Dcm -g
   Sfd_TS_TxDxM2I12R0
   ```

   Hint

   Sfd Module-id usually comes in the form `Sfd_TS_TxDxMxxxxx`

3. An XSL transformation file shall be used be used to transform the EcuC, which shall be created by the user. The module IDs of SFD and other required components are to be set correctly in the transformation file. For a sample implementation of the transformation file, please contact Vector Support (support@vector.com).

1.2 Configuring SFD in CFG5

After the SFD component has been added to the project, the configuration of the component can be done according to **SFD_release_notes_and_user_documentation.pdf [2]** that comes along with the SFD delivery from Elektrobit (see chapter “SFD Configuration”).

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1.3 Configuring Other Dependent Modules in CFG5

Along with the configuration of the SFD component, the modules DCM, DEM, DET, Crypto Stack, NVM, BSWM, which are referenced by the SFD, needs to be configured as described in [2], chapter “Configuration of surrounding modules”. Not all the modules are completely configurable at this point, since the information coming from the SFD SWC descriptions are still missing. However, configurations which are directly referenced by the SFD component can be configured (DCM DIDs and Services, CSM Jobs etc.). If the configurations as per [2] is reported as incorrect/invalid by CFG5, please refer to the OIL excel sheet to make use of possible workarounds. RTE related errors may be ignored at this point.

1.3.1 Deviations in Crypto Configuration

The configuration of RSA and ECDSA signature key verifications shall be done slightly differently in comparison to [2], in order to adapt them for the Vector Crypto solution.

1. The RSA CryptoKeyTypeRef shall be configured to CryptoKeyTypes/Signature_Rsa_Pss_Verify, instead of CryptoKeyType_SIGNATURE_BARRET as mentioned in [2].
   Signature_Rsa_Pss_Verify fits to the CsmPrimitives configuration required for RSA.



Note

The usage of Barrett factor in RSA signature verification is not needed. This is because the Barrett reduction is not implemented in the Vector crypto driver, instead Montgomery reduction is used. The Montgomery reduction is efficient and accepted for the VAG use case.

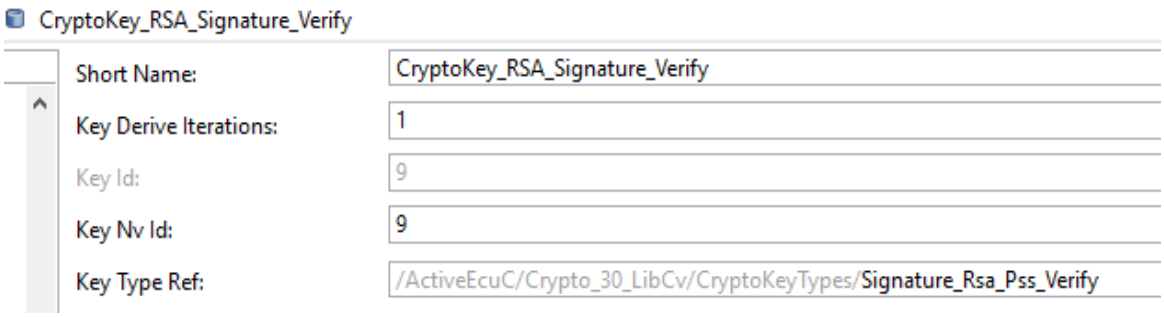


Figure 1-1 CryptoKey_RSA_Signature_Verify

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2. The ECDSA CsmPrimitive shall be configured as shown below

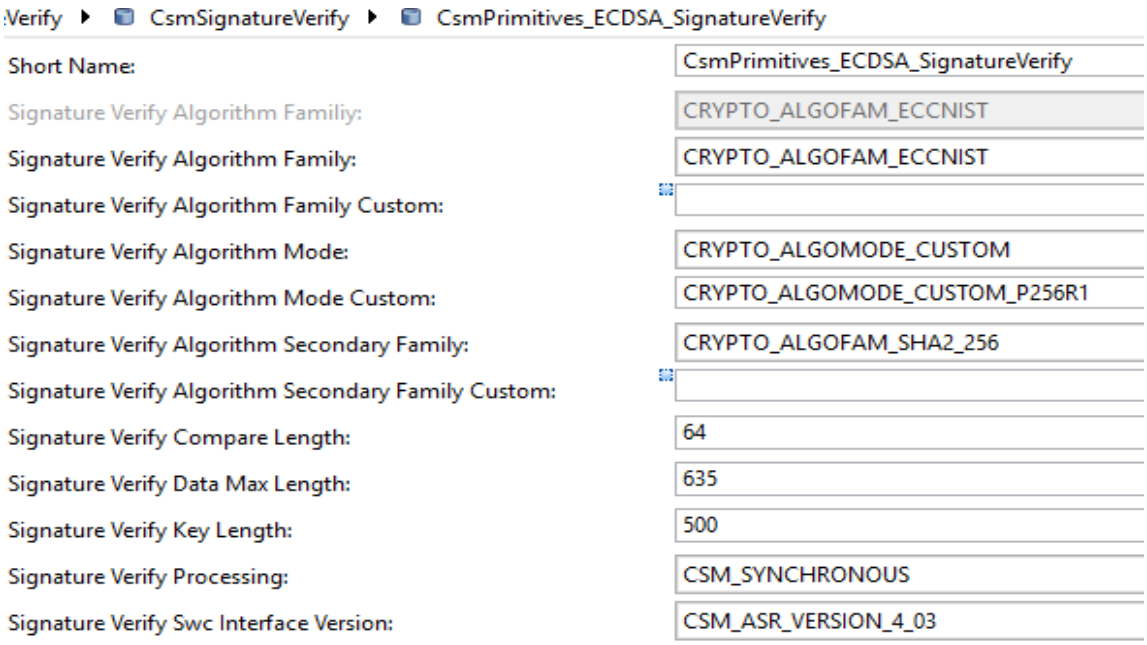


Figure 1-2 ECDSA CsmPrimitive

3. For both signature types, the parameter /MICROSAR/Csm/CsmPrimitives/CsmSignatureVerify/CsmSignatureVerifyConfig/CsmSignatureVerifySwcInterfaceVersion shall be configured to CSM_ASR_VERSION_4_03 to ensure compatibility to CSM interfaces provided by SFD.

   Once all surrounding modules are configured, the external generators for generating the SFD configuration code and SWC descriptions alone shall be invoked. Trying to generate other modules might lead to errors at this point.

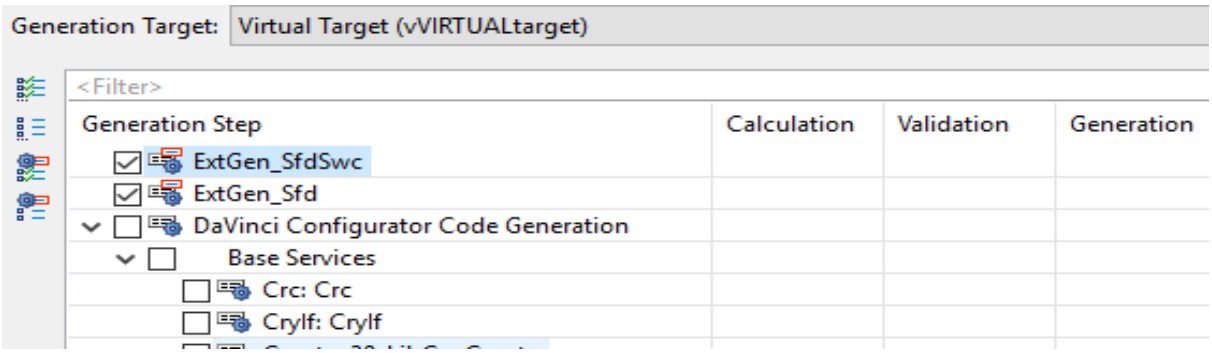


Figure 1-3 Generating SFD SWC and Code

1.4 Importing SFD SWC Descriptions

The generated SWC descriptions (SfdA_ext_interfaces.arxml, SfdA_interface.arxml, SfdA_internal.arxml) shall be imported to the Davinci Developer Workspace of the project.

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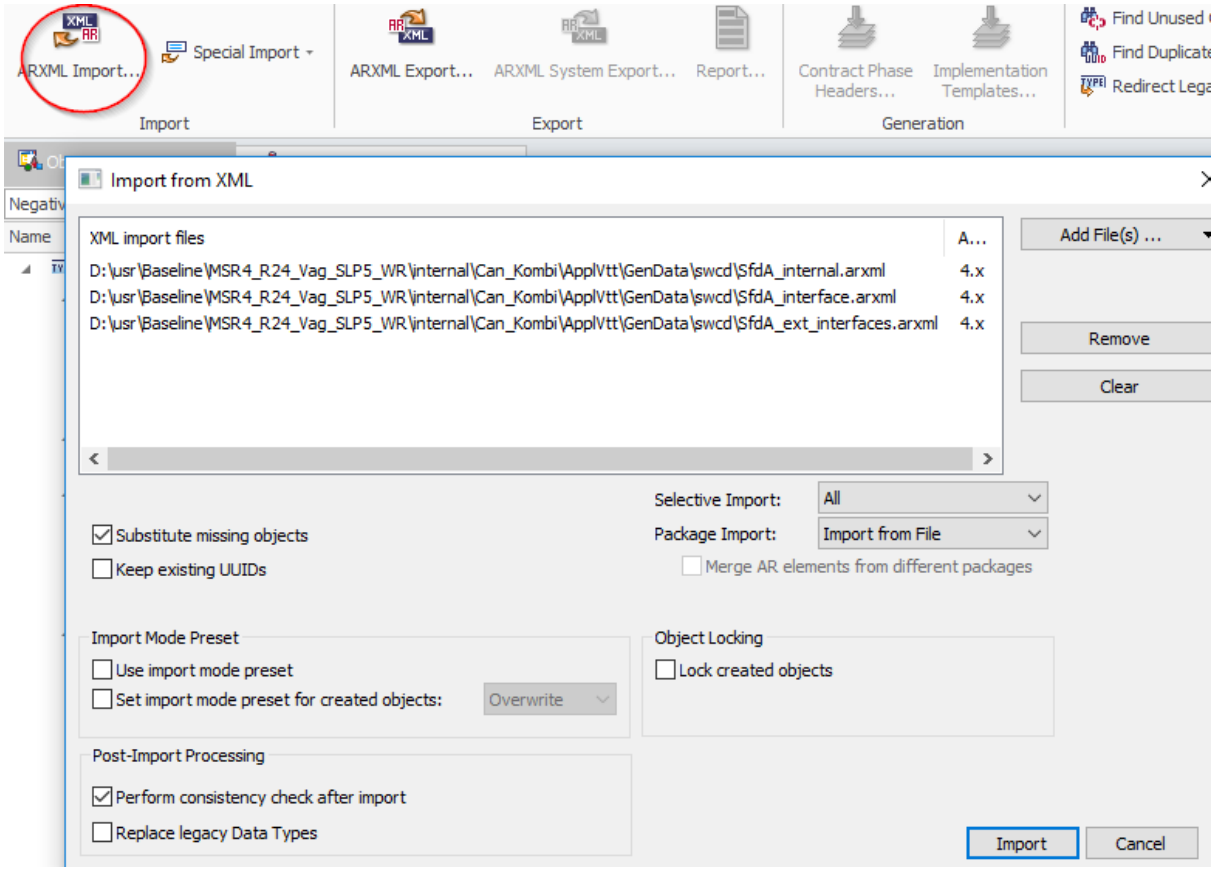


Figure 1-4 SFD ARXML Import

Depending on the used SFD version, several inconsistencies/issues with the imported files might be reported. Please refer to chapter 3 for a list of known issues and workarounds in order to continue with the integration. After this, save the developer workspace. At this point, all the RTE port counterparts from surrounding modules which are expected by various SFD interfaces should be available.

1.5 Configuring SFD RTE Interfaces

After the step above, resynchronize/update the CFG5 project, and add the SFD SWC under application components (/AUTOSAR_SfdA/SfdA/SwComponentTypes/SfdA).

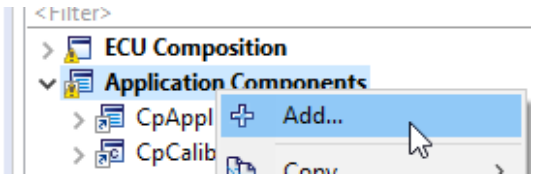


Figure 1-5 Add Application Component

Now it should be possible to connect all the ports provided by SFD to the corresponding ones in dependent modules. Please refer to <Component> RTE Interfaces section of each component in [2], for an overview of interface connections with SFD. If there are still warnings related to data size of arguments in operation prototypes of the interfaces, please refer to chapter 3.

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Port Prototype	Connected Component Prototype	Connected Port Prototype
R_CurMode	BswM	Switch_BswMSwitchPort_SfdAModeNotification
P_RequestRole	BswM	Request_SfdARequestRole
R_CurRole	BswM	Switch_BswMSwitchPort_SfdARoleNotification
R_SignatureVerifyKey	Csm	CsmKey_ECDSA
R_SignatureVerify	Csm	CsmJob_RSA_Signature_Verify_SignatureVerify
R_Hash	Csm	CsmJob_Hash_Hash
P_RS_UnlockRole	Dcm	RoutineServices_DcmDspRoutine_Unlock
P_RS_LockRole	Dcm	RoutineServices_DcmDspRoutine_Lock
P_RS_Reset	Dcm	RoutineServices_DcmDspRoutine_Reset
P_RS_ReqData	Dcm	RoutineServices_DcmDspRoutine_RequestData
P_RS_ReqLog	Dcm	RoutineServices_DcmDspRoutine_RequestLog
P_RS_BeginWriteSecData	Dcm	RoutineServices_DcmDspRoutine_BeginWritingSecureData
P_RS_EndWriteSecData	Dcm	RoutineServices_DcmDspRoutine_EndWritingSecureData
P_RS_ReqConfigProtection	Dcm	RoutineServices_DcmDspRoutine_RequestConfiguration
P_RS_Fingerprint	Dcm	RoutineServices_DcmDspRoutine_CalculateConfigurationStateFingerprint
P_DS_UnlockStatusEcuProtection	Dcm	DataServices_DcmDspData_SFD_Unlock
P_DS_ModeOfProtection	Dcm	DataServices_DcmDspData_SFD_ModeOfProtection
P_DS_Activation_of_Development_Messages_Param_DataRecor	Dcm	DataServices_Activation_of_Development_Messages_Param_DataRecor
P_DS_BAP_Debug_Mode_Release_Param_Relea	Dcm	DataServices_BAP_Debug_Mode_Release_Param_Relea
P_RN_Manufacturer	Dcm	ServiceRequestManufacturerNotification_DcmDslServiceRequestManufacturerNotifi...
R_DiagMonitor	DemSatellite_Appl	Event_SECDIAG_ROLE_UNLOCKED
R_DETService	Det	SFD_DET_Inst
R_PersistentData	NvM	PS_SfdA_PersistentData
R_LogData	NvM	PS_SfdA_LogData
R_ConfigurationFingerprintData	NvM	PS_SfdA_CalcConfData

Figure 1-1-6 Port Connections from SFD to other components

1.6 SfdSetKey Feature

Depending on the chosen handling of the Key used for the signature verification by the setting of the property “Sfd/SfdGeneral/SfdSetKey” additional Steps might need to be considered to align the SFD expectations toward the crypto service manager from Vector. For this purpose, following two options are available.

Option1:
In case this switch is set to FALSE, the customer needs to configure the key manually in the integration code, for example during the startup of the ECU in "BswM_ESH_OnEnterRun".

Example:
In this example the Key used have the Id = 3, where the Modulus and Exponent are fix Id as described in the Crypto-Technical reference and represents a Vector specific Id-Assignment with following charachteristics:

Modulus Id = 160
Exponent Id = 161
SaltLength is an optional parameter which is not used in this example.

```
/* ********** Begin SFD Key Initialization ********** */
/* Values needs to reflect the configuration:
 * CsmKey_RSA_Signature_Verify Key Id = 3
 * Crypto_30_LibCv_Rsa_Modulus Key Element Id =  160
 * Crypto_30_LibCv_Rsa_Modulus Key Element Size =  384
 * Crypto_30_LibCv_Rsa_Pub_Exponent Key Element Id = 161
 * Crypto_30_LibCv_Rsa_Pub_Exponent Key Element Size = 3
 */

/* Set modulus key element according to [QLAH_SFD_1299]*/
retVal = Csm_KeyElementSet(3u, 160u, &SfdKey[9], 384u);
```

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```
if (retVal == E_OK)
{
   /* Set exponent key element */
   retVal = Csm_KeyElementSet(3u, 161u, &SfdKey[395], 3u);
}

if (retVal == E_OK)
{
   /* Check key */
   (void) Csm_KeySetValid(3u);
}

/* ********** End of SFD Key Initialization ********** */
```

Option2:
In case property “Sfd/SfdGeneral/SfdSetKey” is set to TRUE, the API within the SfdA_Crypto.c is used by SfdC_Main_States.c for this purpose. The SFD stick to AUTOSAR standard and use only one KeyElement with ID = 1 which is not reflecting the setting in Vector Crypto-Stack solution. Instead, the KeyElement is split to two keyElements (Modulus and Exponent) with ID 160 respectively 161 like described in the previous example.

Thus, the call of this API will not work in combination with Vector Crypto-stack and needs to be adapted by implementing a wrapper to parse and extracts the necessary KeyElements.

Example:

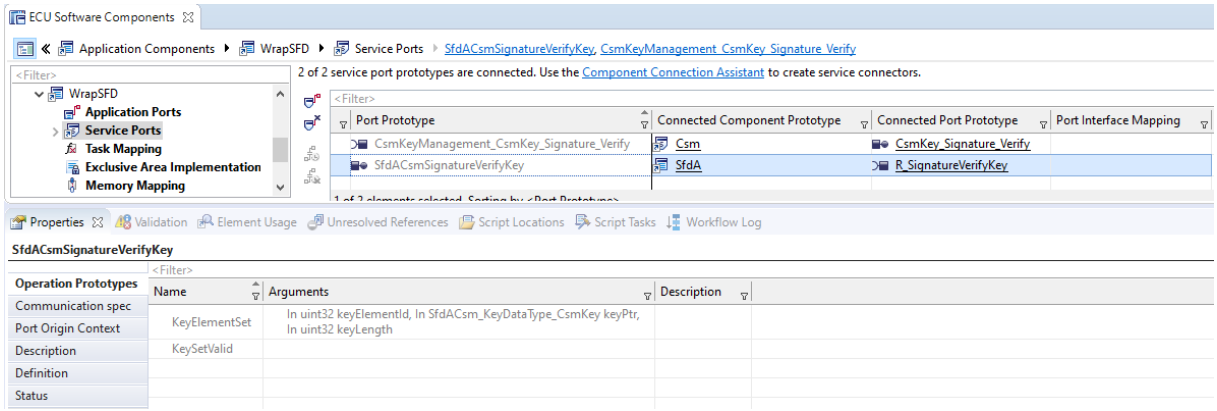


Figure 1-7 example wrapper "WrapSFD" in CFG5

A sample wrapper description and implementation can be requested from vector. Please contact Vector Support (support@vector.com).

1.7 Common Pitfalls

1. **DcmASRServiceAPI-Version**:

EB is using an own annotation to identify the DcmASRServiceAPI-Version. This parameter is not AUTOSAR conform and thus not available in the DCM configuration file (ECUC). The DCM API needed in this case is xx_ReadDataLength. Vector DCM is able to generate the needed API xx_ReadDataLength if UINT8_N is used AND DcmDspDataFixedLength is disabled. Vector DCM is not providing dataType UINT8_DYN, as it is not fully supporting AUTOSAR_43.

EB Generator is generating the API xx_ReadDataLength only if following constraints are met:

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a. DcmASRServiceAPI (=AUTOSAR_40) is available in the ECUC-File AND DcmDspDataFixedLength is disabled AND dataType is UINT8_N.

b. DcmASRServiceAPI (=AUTOSAR_43) AND dataType UINT8_DYN (DcmDspDataFixedLength will be ignored in this case).

Solution:

In the XSL Transformation file, mention the correct value for DcmASRServiceAPI, while changing DcmDemApiVersion(CFG5) to DcmASRServiceAPI(Tresos). Please contact Vector Support for a sample of the XSL Transformation file.

Also, make sure the Property **Dem Api Version** is set as it will be used by the transformation file to replace the needed Tag by EB-Tresos Generator:



Figure 1-8 Dem Api Version

2. DcmDspDataSize vs. DcmDspDataByteSize

For some E2E related services there is the need from SFD to interpret this information and allocate the payload within a DataObject by using some DCM-APIs. Thus the interpretation will fail and the expected APIs are not generated.

Solution:

SFD provides a configuration possibility to switch handling of the generator. Select `Sfd/SfdEndToEndProtectedDIDs/SfdDidDataOffsetType = BITS` to align to our DCM solution `Dcm/DcmConfigSet/DcmDsp/DcmDspData/DcmDspDataSize`.

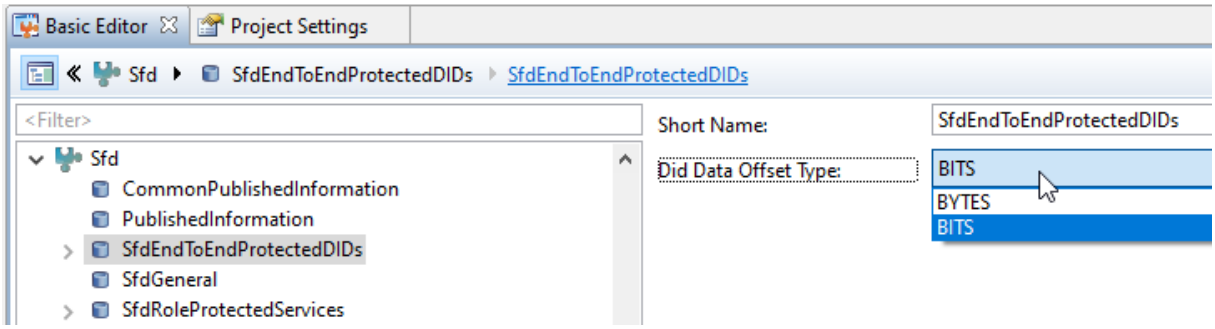


Figure 1-9 Did Data Offset Type

3. Signal length of DataOut_NumAndPvdObj

A common error found in SFD-DCM RTE interface is related to the erroneous configuration of signal length of <OutputSignalName>_DataOut_NumAndPvdObj. This results in an incompatible RTE interface between SFD and DCM.

Solution:

Make sure to configure the length as described in [2], as per the current configuration of the number SFD E2E protected DIDs and SFD role protected services.

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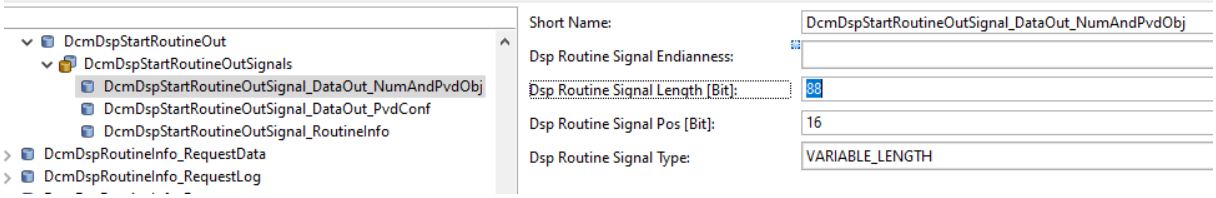


Figure 1-10 Dsp Routine Signal Length Configuration

After all port connections are made, it should finally look like this (there might be additional ports depending on the actual configuration):

1.8 Major Known Issues and Workarounds (as of SFD v2.12)

1. Davinci Developer reports the following invalid references during the SFD SWCD import: SfdANvM_SrcPtrType / SfdANvM_DstPtrType – void. This is because the referenced types(ConstVoidPtr and VoidPtr respectively) are not part of the Autosar_Platform package.

   Workaround:

   Ignore the missing reference during the import and manually change the reference with dtRef_const_VOID for SfdANvM_SrcPtrType and dtRef_VOID for SfdANvM_DstPtrType in Davinci Developer.

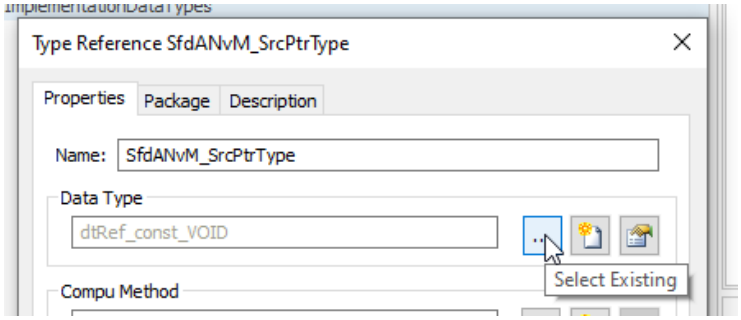


Figure 1-11 Type Reference for SfdANvM_SrcPtrType

2. The current definition of MDG_SfdAModes may result in `SfdA_DeInit()` being called before `SfdA_Init()`. Since UNINIT is the default value of this mode declaration group, `Rte_Start()` activates this mode, triggering SfdA_DeInit(), before `SfdA_Init()` is triggered by BswM mode switch.

   Workaround:

   In Davinci Developer and within the SfdA SWC Design, change the trigger for RE_DeInit has been changed from **On-Entry(to UNINIT)** to **On-Transition from INIT to UNINIT**.

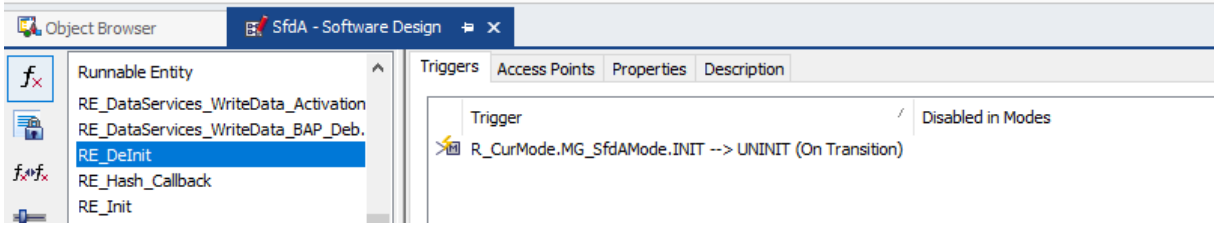


Figure 1-12 Re_DeInit trigger modification

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3. CsmCallbackNotification interface is not compatible with SfdACallbackNotification, the latter has an argument of type Crypto_JobType*. Microsar CSM is AUTOSAR 4.4 compliant, while SFD CSM interface is in compliance with an older version. This gives rise to two issues:

a. The port connection between CsmCallbackNotification and SfdACallbackNotification will result in error RTE51026: Connector Prototype Inconsistent.

Workaround:

The user needs to create a wrapper SWC in Davinci Developer. This SWC shall contain Port Prototypes of both SfdACallbackNotification(from SFD, as client) and CallbackNotification(from CSM, as server). Additionally, a server runnable CallbackNotification shall be added, with an access point for SfdACallbackNotification (Invoked Operation). This is shown below:

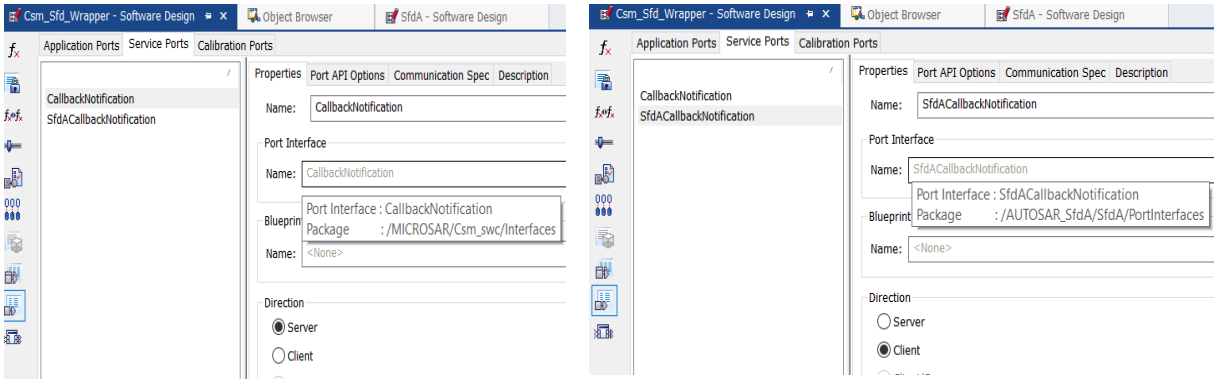


Figure 1-1-14 Port prototypes in Wrapper SWC

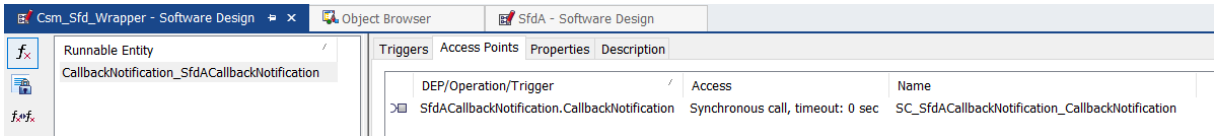


Figure 1-1-13 Access point for SfdACallbackNotification

In CFG5, add the wrapper SWC as application component, and perform the port connections as below:

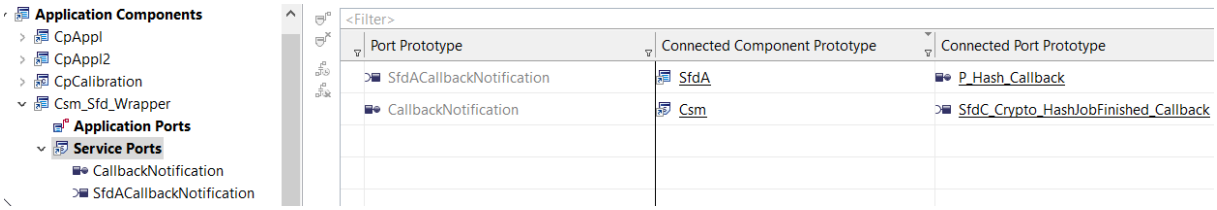


Figure 1-1-15 Port Connections with the Wrapper SWC

The above example shows CsmJob_Hash configured to be asynchronous. If more CSM jobs are configured to be asynchronous, the same procedure must be followed for all related ports and runnables.

b. The Crypto_JobType structure type includes pointer references, and the MICROSAR 4 RTE doesn't support data reference at record element in older MICROSAR releases (it is supported from release R26). This results in error RTE30094 in CFG5.

Workaround:

The Crypto_JobType argument is removed from the CallbackNotification operation prototype of SfdACallbackNotification interface, and Crypto_JobType itself and all the referenced types within it

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(Crypto_JobPrimitiveInfoType, Crypto_JobInfoType, Crypto_JobStateType, Crypto_JobPrimitiveInputOutputType) are removed. This should fix the error RTE30094 in CFG5.

Note

This workaround is applicable only if no CSM jobs are configured to be asynchronous.

For deliveries with releases less release R26 we generally recommend updating to R26 to avoid additional RTE error message RTE30049:

RTE30049	A design constraint is not fulfilled in usage of data reference at data type </AUTOSAR_SfdA/SfdA/ImplementationDataTypes/SfdANvM_DstPtrType>. Relevant constraints: - data reference is allowed to be used only as an input argument of the service client/server port interface Help: - please ensure the restrictions are correctly applied

Alternatively following workaround, which lead to manipulate the SFD description but no influence on the functionality, is available:

Create a „Data Reference“ datatype and name it similar to the Type Reference „SfdANvM_DstPtrType“. The new created datatype shall refer to the Base type “dtRef_VOID” (SfdANvM_SrcPtrType analog):

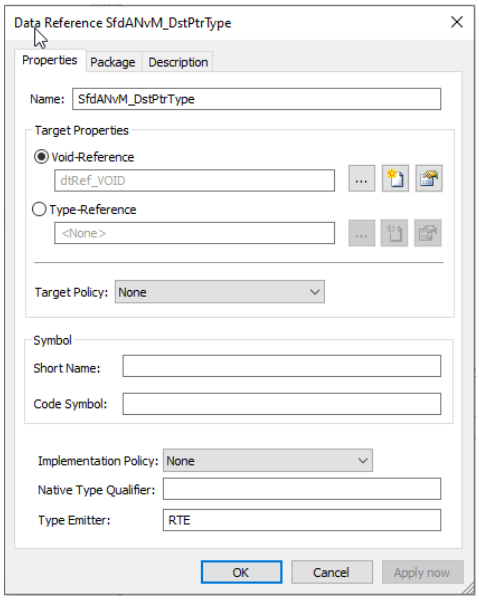


Figure 1-16: Data reference SfdNvM_DstPtrType

The new created datatype can be used in the SFD interface description to fulfil the RTE design constraint

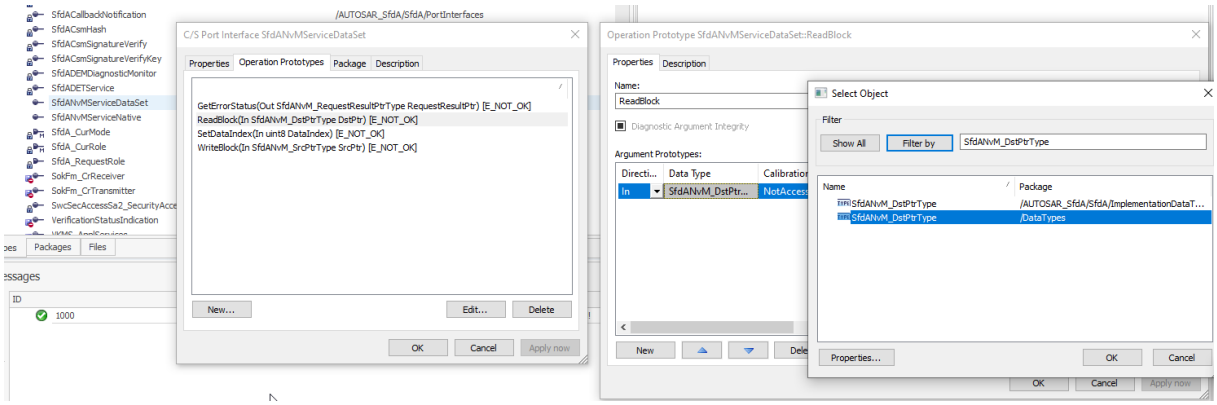


Figure 1-17: usage of SfdNvM_DstPtrType

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2 Additional Resources

1. **AN-ISC-8-1241_Tresos_Plugin_Integration_ DaVinci** Tresos Plugin for Integration in DaVinci Version 1.0.0 in ($SIP_Directory)\Doc\ApplicationNotes\

2. **SFD_release_notes_and_user_documentation** SFD release notes and user documentationModule release 2.x <$SFD_Delivery>\SFD_release_notes_and_user_documentation.pdf

3 Annex

No	Modules	Step	Description	Re-spon-sible	Statu s SFD 2.12	Remarks for SFD 2.12
1	SFD	Configuration	SFD provides no checks if selected DIDs are feasible for SfdEndToEndProtectedDIDs, e.g. they have Read and Write Access configured	EB	open	
2	SFD	SWC Import	DV Developer has the following invalid references: 1) SfdANvM_SrcPtrType / SfdANvM_DstPtrType - void	EB Vector	open	SfdANvM_DstPtrType and SfdANvM_SrcPtrType still needs to be manually adjusted in developer.
3	Crypto	Configuration	According to the EB integration documentation "SFD release notes and user documentation" Version 2.4.0 a CryptoKeyType shall be used to verify signatures to unlock and reset tokens. But the used Name for the KeyType "CryptoKeyType_SIGNATURE_BARRET" is not described in the document. Customers which are using Vector Crypto-Stack are struggeling with the configuration of such a CryptoKeyType! It would be good to extend the EB documentation AND to provide a pre-configuration from Vector to fullfil the SFD requirements related to Crypto-Stack.	Vector EB	open	
4	DCM	Ports	RTE51013 Connector prototype inconsistent. (2 messages) 'No matching operation prototype in any connected port found for operation prototype 'RequestResults' in required port 'RoutineServices_TABROW_UnlocProteOf VehicDiagn' in component prototype 'Dcm'.' 'No matching operation prototype in any connected port found for operation prototype 'Stop' in required port 'RoutineServices_TABROW_UnlocProteOf VehicDiagn' in component prototype 'Dcm'.'	AUDI	open	
5	SFD	Runtime	The current definition of MDG_SfdAModes results in SfdA_DeInit() being called before SfdA_Init(). Since UNINIT is the default value of this mode declaration group, Rte_Start() activates this mode, triggering SfdA_DeInit() (Requirement [SWS_Rte_06766]), before SfdA_Init() is triggered by BswM mode switch.	EB	open	As a workaround, the trigger for RE_DeInit has been changed from On-Entry(to UNINIT) to On-Transition from INIT to UNINIT.

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